

Semester: August 2021 - November 2021 Examination: In Semester Examination		
Programme code: 01/04 Programme: B. Tech Computer Engineering	Class: SY	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department/Section/Center: COMP/IT	
Course Code: 116U01L302	Name of the Course: Data Structures	

Questi on No.		Max. Marks
Q1.1	Reverse Polish notation is the other name of (a) Infix expression (b) Prefix expression (c) Postfix expression (d) Algebraic expression	01
Q1.2	Which type of linked list does not store NULL in next field? (a) Singly linked list (b) Circular linked list (c) Doubly linked list (d) All of these	01
Q1.3	Which of the following is not an application of priority queue? a) To do list b) Waiting queue in hospital c) Undo operation in text editors d) Resource sharing among users	01
Q1.4	The data structure required to check whether an expression contains a balanced parenthesis is? a) Stack b) Queue c) Array d) Tree	01
Q1.5	The operator definition of ADT must have a) Abstract, return data types, operation name , input parameters, precondition , post condition a) Operation name , input parameters, precondition , post condition, value definition c) Operation name , precondition , post condition d) Value definition , abstract, operation name , input parameters, precondition , post condition	01
Q1.6	Linked lists are not suitable for the implementation of a) Waiting queue b) Bubble sort	01

	c) Polynomial manipulation d) Family Tree	
Q1.7	Which of the following application makes use of a circular linked list? a) Undo operation in a text editor b) Recursive function calls c) Allocating CPU to processes d) Checking string for palindrome	01
Q1.8	What does 'stack underflow' refer to? a) Accessing item from an undefined stack b) Adding items to a full stack c) Removing items from an empty stack d) Index out of bounds exception	01
Q1.9	Which of the following is true about the characteristics of abstract data types? i) It exports a type. ii) It exports a set of operations. a) True, True b) True, False c) False , True d) False , False	01
Q1.10	Choose correct output for the following sequence of operations :- push(5), push(8),pop,push(2),push(5),pop,pop,pop,push(1),pop a) 85521 b) 82551 c) 81255 d) 85251	01
Q2 (a)	How do you use stack for checking if parenthesis in an expression are balanced or not? Consider a parenthesis sequence : {[([)]]}]. Show how the parenthesis would be checked as an application of stack. Show the step by step working of the same. OR What is recursion? How the stack will be used in recursive function calls? Explain the concept with a scenario of factorial computation function that uses recursion.	05
Q2 (b)	Explain Deque concept. Write the insertion process in a doubly ended queue, consider all possible cases. Assume that the Deque doesn't have any restrictions on input or output. OR Discuss benefits of Circular queue over normal queue? Explain the Circular Queue concept with suitable example.	05
Q (3)	Differentiate between Single linked list V/S Double linked list . Explain and write algorithms for the following doubly linked list(DLL) operations. a) Insert_begin b) Insert_before c) Delete_after d) Delete_end	(2+8)